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MedCLIPSeg: Probabilistic Vision-Language Adaptation for Data-Efficient and Generalizable Medical Image Segmentation

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https://tahakoleilat.github.io/MedCLIPSeg

Abstract

Medical image segmentation remains challenging due to limited annotations for training, ambiguous anatomical features, and domain shifts. While vision-language models such as CLIP offer strong cross-modal representations, their potential for dense, text-guided medical image segmentation remains underexplored. We present `MedCLIPSeg`*, a novel framework that adapts CLIP for robust, data-efficient, and uncertainty-aware medical image segmentation. Our approach leverages patch-level CLIP embeddings through probabilistic cross-modal attention, enabling bidirectional interaction between image and text tokens and explicit modeling of predictive uncertainty. Together with a soft patch-level contrastive loss that encourages more nuanced semantic learning across diverse textual prompts,* `MedCLIPSeg` *effectively improves data efficiency and domain generalizability. Extensive experiments across 16 datasets spanning five imaging modalities and six organs demonstrate that* `MedCLIPSeg` *outperforms prior methods in accuracy, efficiency, and robustness, while providing interpretable uncertainty maps that highlight local reliability of segmentation results. This work demonstrates the potential of probabilistic vision-language modeling for text-driven medical image segmentation.*

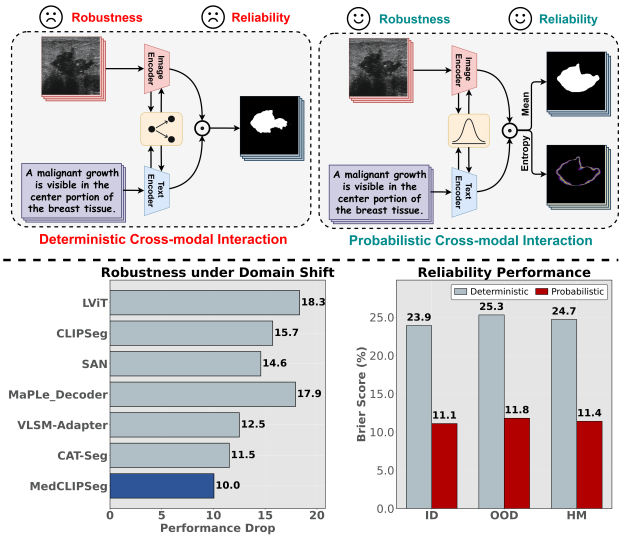


Figure 1. ***(Top)***: **Comparison between deterministic and probabilistic cross-modal fusion techniques in CLIP adaptation for text-driven segmentation.** Probabilistic formulation models variability in visual–textual representations as distributions, enabling more robust feature alignment. ***(Bottom)***: **Robustness** and **Reliability** plots over ID and OOD data show improved generalization, with smaller out-of-domain performance drops and better calibration of predicted confidence, reflected by lower Brier scores.

1. Introduction

Accurate and trustworthy medical image segmentation remains a cornerstone for diagnosis, treatment planning, and quantitative clinical follow-up. Yet progress is often constrained by three persistent obstacles. First, *expert annotations* of segmentation ground truths are expensive and often inconsistent across raters, restricting the quality of supervised learning. Second, lesions and organs can exhibit *ambiguous boundaries* due to gradual intensity transitions or partial-volume effects that make clear decision-making difficult. Third, common *domain shifts* in scans due to variations in scanners, acquisition protocols, and patient populations cause models trained on limited in-distribution (ID) data to fail when exposed to out-of-distribution (OOD) conditions. These issues expose an urgent need for segmentation systems that are simultaneously data-efficient, uncertainty-aware, and generalizable across domains.

In medical image segmentation, U-Net and its variants [34, 63] have driven major advances, by exploiting convolu-

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tional neural network (CNN)'s inductive biases for efficient feature learning and/or Vision Transformer (ViT) architectures [9, 11, 30] for long-range dependencies. Despite their success, these models depend on extensive pixel-wise supervision and most operate deterministically. Prior work shows that such segmentation networks, including U-Nets, are systematically *over-confident*, particularly for out-of-distribution inputs and those with fuzzy tissue boundaries [3, 74], yielding unreliable segmentation results without warning mechanisms. This highlights the issues in conventional feature learning with deterministic approaches, which fail to properly account for ambiguity or local disagreement among features. Learning methods that can adaptively weigh evidence and/or modulate attention based on contextual reliability of data distribution would be hopeful to address this [53]. Additionally, alternative methods that move beyond traditional fully supervised approaches to mitigate the heavy reliance on expert training data while enhancing generalizability and interactability are highly desirable.

By aligning images and texts through large-scale contrastive pretraining, vision–language models (VLMs) such as CLIP [60] and its biomedical variants [43, 76] offer a promising paradigm towards more label-efficient and generalizable medical image segmentation with the potential for intuitive natural language-driven user-interaction. Notably, recent studies show that CLIP's patch tokens can encode spatial semantics [25, 82], suggesting its capacity for dense localization even without pixel-level supervision. Yet, in medical domain, where more subtle visual differences and fine-grained descriptions limit multi-modal alignment [80], the dense localization capability in VLMs remains weaker [23], but can be potentially strengthened through *more nuanced image–text mapping* [48] and *tailored, task-specific representation adaptation* without degrading generalization [32]. Practically, clinical descriptions are far easier to obtain than pixel-level masks, making VLMs appealing in low-data regimes, where textual supervision compensates for limited annotations [47]. While most methods emphasize prompt learning, decoder tuning, or unidirectional text-to-vision modulation [50, 59, 73], *deep cross-modal fusion* was shown to better support CLIP adaptation and spatial grounding [42, 77]. Yet, CLIP models with deterministic representations still remain over-confident on OOD data [54], motivating probabilistic CLIP formulation [15], which remains underexplored for medical image segmentation.

In this context, a key question emerges: *how can we formulate cross-modal attention to be uncertainty-aware for CLIP-based segmentation?* This is particularly relevant in practical medical AI adoption, where model credence and transparency are crucial, while over-confident deterministic AI models fail to meet the needs. Building on existing insights, we propose `MedCLIPSeg`, a text-driven medical image segmentation framework that adapts CLIP with probabilistic, bidirectional vision–language representation fusion. Its core component, the *Probabilistic Vision–Language (`PVL`) adapter* learns confidence-weighted attention between image patches and text tokens within CLIP's multiple deep encoding layers. Confidence-aware attention scores based on variational modeling of *Keys* reduce over-confidence, and Monte Carlo sampling of *Value* distributions yields both mean segmentation masks and associated pixel-level uncertainty maps for user interpretation. Specifically, our probabilistic modeling of the *Keys* and *Values* in cross-modal attention naturally captures *aleatoric uncertainty* from ambiguous image features and *epistemic uncertainty* arising from unseen domains, leading to better accuracy, improved calibration, and enhanced robustness, in line with [29, 40]. As illustrated in Fig. 1, in contrast to `MedCLIPSeg`'s probabilistic adaptation, the deterministic variant leads to inaccurate segmentation and over-confidence that manifests as poor calibration and significant performance drops under domain shift. Furthermore, to maintain data efficiency, we preserve CLIP's pretrained encoders and introduce a soft patch-level contrastive loss that refines image–text alignment for dense prediction under limited supervision. In summary, our main contributions include:

1. **Bidirectional representation-level fusion** that enhances data-efficiency and robustness through novel vision-language interaction adapters while preserving CLIP's parameters, guided by a soft contrastive loss.
2. **Probabilistic cross-modal attention** with variational *Key* and *Value* formulation to enable uncertainty-aware learning to improve accuracy and generalizability.
3. **Pixel-level uncertainty maps** by sampling *Values* from learnt probability distributions in VLM attentions to offer intuitive reliability visualization for clinical review.
4. **Comprehensive evaluation** against SOTA methods for medical image segmentation on five modalities and six organs, assessing data efficiency, domain generalizability, and model sub-component performance to provide insights into the proposed framework.

2. Related Work

Medical Image Segmentation: Medical image segmentation has traditionally relied on vision-only architectures. CNNs established the foundation, with U-Net [63] introducing the long skip connections, inspiring UNet++ [85], Attention U-Net [56], and nnUNet [34]. Other variants, including the DeepLab series [12], improved multi-scale context modeling through dilated convolutions and pyramid pooling. The advent of ViTs further advanced segmentation by incorporating long-range dependencies. TransUNet [11] fused CNN backbones with Transformer encoders, while Swin-UNet [9] improved efficiency using shifted windows. Building on these designs, hybrid models such as

HiFormer [31] and UNETR [30] demonstrated strong performance across multiple benchmarks. Despite these advances, vision-only approaches often rely heavily on low-level appearance features and show limited robustness to domain shifts across scanners and imaging protocols. This motivates the use of high-level semantic priors, particularly via cross-modal learning to improve generalizability.

Vision-Language Models: Vision-language models have gained interest in biomedical domains. As CLIP [60] and ALIGN [38] demonstrated strong zero-shot transfer across different visual tasks, they inspired biomedical variants such as BiomedCLIP [76], PubMedCLIP [21], and UniMedCLIP [43], which leverage clinical image-text corpora for domain-specific learning. While these models provide robust global alignment, they often require further adaptation to capture the fine-grained semantics of anatomy and pathology. Typical parameter-efficient adaptation techniques include prompt tuning (e.g., CoOp [84], CoCoOp [83], and MaPLe [42]) and low-rank-based model updates (e.g., CLIP-LoRA [75] and CLIP-SVD [49]). Tailored for biomedical visions, methods such as DCPL [10] and BiomedCoOp [47] incorporate domain priors and knowledge distillation to enhance adaptation under limited supervision. More recently, probabilistic fine-tuning frameworks such as CLAP4CLIP [37] and ProbVLM [67] incorporate embedding uncertainty to better manage the many-to-one mappings between vision and language, improving calibration and cross-domain generalization. Despite these advances, most biomedical VLMs focus on classification or retrieval, with limited exploration of spatially grounded tasks, such as segmentation, which is more challenging.

Prompt-based Segmentation: For natural images, only a few methods extend VLMs to image segmentation. CLIPSeg [51] and CRIS [69] append lightweight decoders to frozen CLIP encoders, while LAVT [73] fuses textual embeddings into Transformer layers through cross-attention, modulating visual features throughout the encoder. Further approaches such as DenseCLIP [61], ZegCLIP [86], and SAN [72] enhance fine-grained localisation, while the recent CAT-Seg [16] stands out with strong state-of-the-art performance in open-vocabulary segmentation. On the other hand, the Segment Anything Model (SAM) [45] introduced a promptable foundation for general-purpose segmentation but lacks explicit natural language interaction and conditioning, and remains limited to drawing-based prompts. Although effective for natural images, these methods struggle in biomedical contexts, where images lack contextual diversity, exhibit high inter-class similarity, and feature ambiguous boundaries. On the other hand, medical adaptations, like LViT [50] and Ariadne's Thread [81], integrated BERT textual embeddings into ViT architectures. MedSAM [52] and recent prompt-based work [46, 48, 62, 64, 70] still rely on geometric prompts at different intermediate stages, introducing potential instability. More recently, few-shot medical image segmentation frameworks, such as UniverSeg [7], MultiverSeg [71], and Iris [26], leverage a small support set of image–label pairs to segment unseen classes and modalities without additional training. BiomedParse [78] further explores structured knowledge parsing across modalities through the use of natural language. However, methods for adapting VLMs such as CLIP for dense biomedical prediction remain limited. Poudel *et al.* [59] applied CLIPSeg and CRIS with frozen CLIP encoders and new decoders, but domain-specific models such as BiomedCLIP [76] offered no gains, highlighting the weakness of naïve adaptations. VLSM-Adapter [19] offers parameter-efficient VLM adaptation, and CausalCLIPSeg [13] introduces multi-modal causal adaptations, but evaluations on multiple datasets are not provided, and we show its weakness in domain generalization (Table 2). Unlike existing approaches, our method preserves CLIP's pretrained parameters while introducing probabilistic, bidirectional vision-language fusion to enhance robustness and clinical reliability in biomedical segmentation.

3. Methodology

Our `MedCLIPSeg` framework is presented in Fig. 2.

3.1. CLIP Overview

Our text-driven segmentation framework builds upon the CLIP architecture [60], which comprises two Transformer-based encoders: a vision encoder $\boldsymbol{E_v}$ and a text encoder $\boldsymbol{E_t}$. They encode images and text into a shared D-dimensional space for cross-modal alignment. For a batch of B RGB images, the input is represented as $\boldsymbol{X_v} \in \mathbb{R}^{B\times 3\times H\times W}$. The vision encoder first partitions each image into P non-overlapping patches, projects each to a D-dimensional embedding, and prepends a learnable `[CLS]` token. This results in a sequence of $(P+1)$ tokens:

$$\boldsymbol{Z_v} = \boldsymbol{E_v}(\boldsymbol{X_v}) \in \mathbb{R}^{B\times(P+1)\times D} \tag{1}$$

The `[CLS]` token output, $\boldsymbol{Z_v}^{\texttt{[CLS]}} \in \mathbb{R}^{B\times D}$, serves as the global image representation in the shared CLIP space. Notably, prior works [25, 82] have shown that, although CLIP's objective aligns only the `[CLS]` tokens, the contrastive vision-language pre-training implicitly shapes the patch token representations to encode semantically meaningful spatial features with emerging textual correlation.
For the textual branch, tokenized prompts $\boldsymbol{X_t} \in \mathbb{R}^{B\times L}$ of length L are embedded into D-dimensional vectors and processed by the text encoder. Unlike the vision encoder's use of a `[CLS]` token, CLIP takes the output at the `[EOS]` position as the global text representation:

$$\boldsymbol{Z_t} = \boldsymbol{E_t}(\boldsymbol{X_t}) \in \mathbb{R}^{B\times L\times D} \tag{2}$$

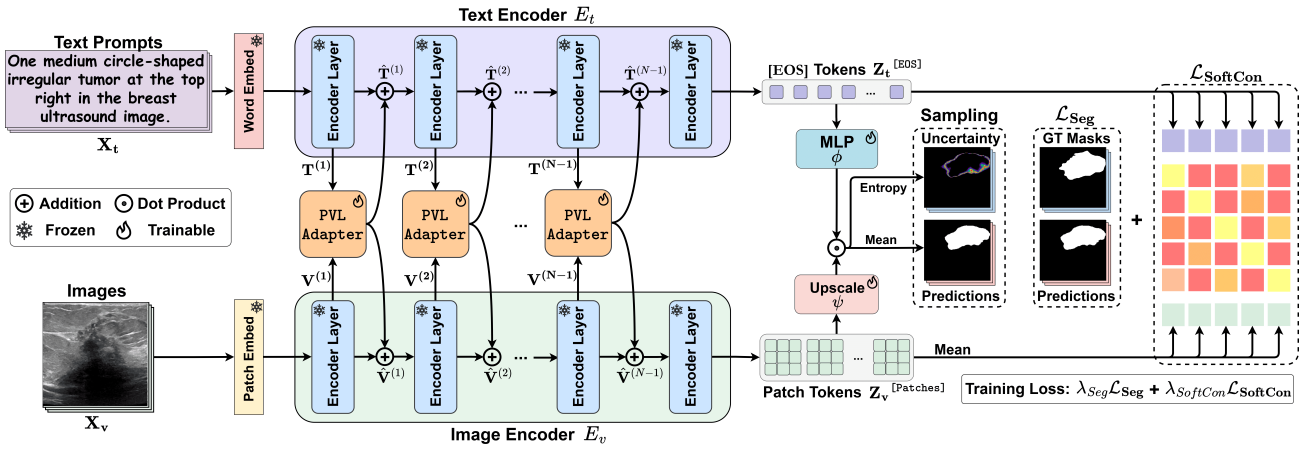


Figure 2. Overview of the proposed `MedCLIPSeg` framework for text-driven medical image segmentation. The model extends CLIP with vision and language encoders connected via `PVL Adapters`, which perform confidence-weighted image–text fusion at multiple deep layers. Segmentation and uncertainty maps arise from the *mean* and *entropy* of posterior samples, with a soft patch-level contrastive loss.

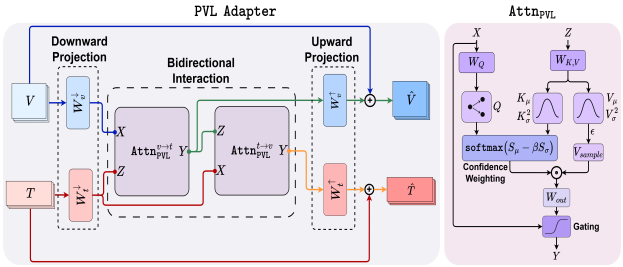


Figure 3. Illustrations of **`PVL Adapter`** and **$\texttt{Attn}_{\texttt{PVL}}$**.

Here, the `[EOS]` output $\boldsymbol{Z}_t^{[\text{EOS}]} \in \mathbb{R}^{B\times D}$ acts as the compact textual embedding in the joint space. The global embeddings enable cross-modal similarity, while the patch tokens from $\boldsymbol{Z_v}$ preserve fine-grained spatial information. In our framework, the global text embedding $\boldsymbol{Z}_t^{[\text{EOS}]}$ is used as a query and compared via dot product with each vision patch token, producing segmentation logits that guide segmentation according to the natural language prompt.

3.2. Probabilistic Multi-modal Adaptation

To enable efficient and confidence-aware multimodal fusion-based CLIP adaptation for biomedical segmentation, we propose the *Probabilistic Vision-Language Adapter* (`PVL Adapter`), a novel probabilistic framework that bridges CLIP's vision and language encoders at the representation level. Each `PVL` module performs bidirectional, probabilistic interaction between image and text tokens. The architecture of `PVL Adapter` is depicted in Fig. 3.

Downward Projection: Given visual tokens $\mathbf{V}^{(n)} \in \mathbb{R}^{B\times T_v\times D_v}$ and text tokens $\mathbf{T}^{(n)} \in \mathbb{R}^{B\times T_t\times D_t}$ at Layer n with total layers amounting to N, the `PVL` adapter first projects both modalities to a shared lower-dimensional space D_s with $W_v^{\downarrow(n)} \in \mathbb{R}^{D_v\times D_s}$ and $W_t^{\downarrow(n)} \in \mathbb{R}^{D_t\times D_s}$:

$$\mathbf{v}^{(n)} = \mathbf{V}^{(n)}W_v^{\downarrow(n)}, \quad \mathbf{t}^{(n)} = \mathbf{T}^{(n)}W_t^{\downarrow(n)} \tag{3}$$

QKV parameterization: Inspired by [40], we extend the standard attention formulation to incorporate uncertainty in both *Keys* and *Values* by modeling them as probability distributions with learnable means and variances. These variances represent data ambiguity, allowing the model to encode inherent noise in the input representations. This probabilistic design enables the attention module to down-weight uncertain tokens and sample value representations for stochastic modeling. This attention module $\texttt{Attn}_{\texttt{PVL}}$ takes as input a query sequence X and a context sequence Z and outputs a fused output Y as $Y = \texttt{Attn}_{\texttt{PVL}}(X, Z)$. The input query and context sequences are transformed using the Query (Q), Key (K), Value (V), and Output (O) projection matrices as follows:

$$Q = X \cdot W_Q \tag{4}$$

where $X \in \mathbb{R}^{B\times T_q\times D_s}$ is the input query sequence, $W_Q \in \mathbb{R}^{D_s\times D_a}$ is the learnable projection matrix, and Q is the transformed query to compute the attention score.
Keys and Values are projected into both a mean and a log-variance representation:

$$[K_\mu,\, K_{\log\sigma^2}] = Z \cdot W_K, \quad [V_\mu,\, V_{\log\sigma^2}] = Z \cdot W_V \tag{5}$$

where $Z \in \mathbb{R}^{B\times T_k\times D_s}$ is the context sequence and $W_K \in \mathbb{R}^{D_s\times 2D_a}$ produces two D_a-dimensional splits for mean and log-variances.

To convert the predicted log-variances into original variance values, we avoid the numerically unstable $\exp(\cdot)$ and instead use the $\texttt{softplus}\ \zeta(.)$ activation, which is smoother and less prone to instabilities:

$$K_\sigma^2 = \zeta(K_{\log \sigma^2}), \quad V_\sigma^2 = \zeta(V_{\log \sigma^2}) \tag{6}$$

Here, K_σ^2 and V_σ^2 represent the variance terms for the *Key* and *Value* distributions.

Confidence-weighted attention: Our proposed attention score considers two terms: a mean similarity S_μ and a variance-based confidence penalty S_σ^2. Each score S_{ij} corresponds to the dot product between a deterministic query vector Q_i and a probabilistic key $K_j \sim \mathcal{N}(K_{\mu,j}, K_{\sigma,j}^2)$, assuming feature-wise independence. This yields a probabilistic attention score with mean and variance given by:

$$S_\mu = \frac{QK_\mu^\top}{\sqrt{D_a}}, \quad S_\sigma^2 = \frac{Q^{\circ 2}(K_\sigma^2)^\top}{D_a}, \tag{7}$$

where $Q^{\circ 2}$ denotes the element-wise square of Q. The variance term quantifies how key uncertainty interacts with the magnitude of query features, allowing uncertain tokens to be adaptively downweighted. The final attention weights are computed as:

$$A = \{A_{ij}\} = \texttt{softmax}\big(S_\mu - \beta S_\sigma\big)\ with \tag{8}$$

$$A_{ij} = \frac{\exp\big(S_{\mu,ij}\big)/\omega_{ij}}{\sum_r \exp\big(S_{\mu,ir}\big)/\omega_{ir}}, \quad \omega_{ij} = \exp\big(\beta\, S_{\sigma,ij}\big) \tag{9}$$

where $\beta = 2.35$ (indicates the full width at half maximum of a Gaussian distribution) scales the confidence penalty, down-weighting overconfident attention responses. This dynamic probabilistic weighting allows the model to emphasize reliable evidence and downweight uncertain signals, thereby regularizing feature learning and improving *out-of-distribution generalization*, which is crucial for medical data with large domain variability. Conceptually, Eq. 8 can be interpreted as a *variance-aware extension* of the standard attention mechanism, where attention scores are weighted by an input-dependent uncertainty term (Eq. 9); when $\beta = 0$, the formulation naturally reduces to the conventional deterministic attention.

Value sampling: To model uncertainty in the *Value* representations, we draw samples from their learned probability distribution $\mathcal{N}(V_\mu, V_\sigma^2)$. Each sample is obtained via the reparameterization trick:

$$V_{\text{sample}} = V_\mu + \epsilon \odot V_\sigma, \quad \epsilon \sim \mathcal{N}(0, I). \tag{10}$$

where $\odot$ denotes the Hadamard product.

During training, we adopt a stochastic regime by sampling only once to compute the attended output as:

$$O = A \cdot V_{\text{sample}}. \tag{11}$$

At test time, we perform multiple stochastic forward passes to sample from the model's approximate posterior. The learned variances within the $\texttt{PVL Adapters}$ capture aleatoric uncertainty, while Monte Carlo sampling accounts for epistemic uncertainty arising from model variability. The predictive entropy across sampled outputs quantifies total uncertainty and overall confidence. Empirically, we found 30 forward passes sufficient to obtain stable uncertainty estimates.

Residual gating: Directly relying on attended features from $\texttt{PVL Adapters}$ can introduce instability early in the training schedule, when attention responses are still noisy. The residual gate mitigates this by controlling how much new information is incorporated, gradually increasing reliance on attended features as cross-modal alignment and model confidence improve, leading to smoother optimization and more reliable fusion:

$$O_{\text{proj}} = O \cdot W_{\text{out}}, \tag{12}$$

where $W_{\text{out}} \in \mathbb{R}^{D_a \times D_s}$ is the output projection matrix.

Then we apply a learnable gate $g \in [0, 1]$:

$$Y = g \odot O_{\text{proj}} + (1 - g) \odot X, \tag{13}$$

where scalar g is initialized to a balanced weighting through $\texttt{sigmoid}(0)$, providing equal emphasis on the original query X and the attended output at the start of training.

Bidirectional Interaction: A two-way Transformer layer performs mutual updates via $\texttt{Attn}_{\texttt{PVL}}$, enabling visual and textual features to mutually refine each other for stronger cross-modal alignment and contextual consistency:

$$\mathbf{v}'^{(n)} = \texttt{Attn}_{\texttt{PVL}}^{v \to t}(\mathbf{v}^{(n)}, \mathbf{t}^{(n)}) \tag{14}$$

$$\mathbf{t}'^{(n)} = \texttt{Attn}_{\texttt{PVL}}^{t \to v}(\mathbf{t}^{(n)}, \mathbf{v}'^{(n)}) \tag{15}$$

Upward Projection: Finally, the fused features are projected back to their original dimensions with $W_v^{\uparrow(n)} \in \mathbb{R}^{D_s \times D_v}$ and $W_t^{\uparrow(n)} \in \mathbb{R}^{D_s \times D_t}$ with residual connections applied:

$$\hat{\mathbf{V}}^{(n)} = \mathbf{V}^{(n)} + \mathbf{v}'^{(n)} W_v^{\uparrow(n)} \tag{16}$$

$$\hat{\mathbf{T}}^{(n)} = \mathbf{T}^{(n)} + \mathbf{t}'^{(n)} W_t^{\uparrow(n)} \tag{17}$$

The $\texttt{PVL Adapters}$ is applied at multiple encoding CLIP layers to refine joint representations.

3.3. Segmentation via Pixel-Text Similarity

After the final fusion layer, the text $\texttt{[EOS]}$ token embedding and the visual patch are L2-normalized. The visual patch tokens are then upscaled via a learned block ψ, while

a lightweight MLP mask head ϕ maps $\boldsymbol{Z}_{\boldsymbol{t}}^{[\texttt{EOS}]}$ to a compatible embedding:

$$\tilde{\mathbf{V}} = \psi(\boldsymbol{Z}_{\boldsymbol{v}}^{[\texttt{Patches}]}), \quad \tilde{\mathbf{t}} = \phi(\boldsymbol{Z}_{\boldsymbol{t}}^{[\texttt{EOS}]}) \tag{18}$$

The segmentation logits $\mathbf{M} \in \mathbb{R}^{B\times H\times W}$ are computed by a simple dot product followed by bilinear interpolation:

$$\mathbf{M} = \texttt{Upsample}_{H\times W}(\tilde{\mathbf{V}} \cdot \tilde{\mathbf{t}}^{\top}) \tag{19}$$

3.4. Soft Patch-level Contrastive Loss

Building on CLIP's ability to capture rich image–text relationships, we extend it to handle diverse descriptions with spatial context for medical images. Since a single caption may mention multiple anatomical regions, global alignment alone can be insufficient for fine-grained correspondence. To address this, we average image patch embeddings into stable regional representations that preserve local visual semantics while reducing token-level noise. This enables more accurate text–region associations and consistent segmentation performance across heterogeneous anatomy. Specifically, we align L2-normalized *Average-pooled* visual patch embeddings $p_v = \bar{\boldsymbol{Z}}_{\boldsymbol{v}}^{[\texttt{Patches}]}$ with text embeddings $p_t = \boldsymbol{Z}_{\boldsymbol{t}}^{[\texttt{EOS}]}$ via a *soft contrastive loss*, computing text-to-image similarity logit $\mathrm{P}^{v\to t}$ and image-to-text counterpart $\mathrm{P}^{t\to v}$ across all batch pairs.

Since text prompts within a batch can be similar, hard targets are replaced with soft targets derived from text similarities:

$$\mathrm{G} = \texttt{softmax}\left(\frac{p_t \cdot p_t^{\top}}{\tau}\right), \tag{20}$$

where τ is a temperature parameter set to 0.2. The soft cross-entropy loss is defined in the following, and the final contrastive loss averages both directions:

$$\mathcal{L}_{\text{soft}}(\mathrm{P}, \mathrm{G}) = -\frac{1}{B}\sum_i \sum_j \mathrm{G}_{ij} \log(\texttt{softmax}(\mathrm{P}_i)_j). \tag{21}$$

$$\mathcal{L}_{\text{SoftCon}} = \frac{1}{2}\left(\mathcal{L}_{\text{soft}}(\mathrm{P}^{t\to v}, \mathrm{G}) + \mathcal{L}_{\text{soft}}(\mathrm{P}^{v\to t}, \mathrm{G}^{\top})\right). \tag{22}$$

The overall training loss combines segmentation (Dice+BCE, equal weights) and contrastive objectives:

$$\mathcal{L} = \lambda_{\text{Seg}}\,\mathcal{L}_{\text{Seg}} + \lambda_{\text{SoftCon}}\,\mathcal{L}_{\text{SoftCon}}. \tag{23}$$

where $\lambda_{\text{Seg}}, \lambda_{\text{SoftCon}}$ control the relative importance of each loss term, set to 0.5 and 0.1, respectively.

4. Experiments and Results

4.1. Experimental Setup

Data Efficiency: We evaluate `MedCLIPSeg` under varying levels of training data to assess data-efficiency and scalability. Models are trained with 10%, 25%, and 50% of the data to measure performance under limited supervision. The fully supervised setting uses all training examples with their pixel-level and textual annotations, serving as an upper-bound reference.

Domain Generalization: To assess out-of-distribution (OOD) generalization, models are trained on one in-distribution (ID) source dataset, fully supervised, and directly tested on unseen target datasets without fine-tuning. This evaluates robustness to domain shifts in image acquisition, clinical sites, and patient populations.

Table 1. **Data-efficiency evaluation:** This table reports the average DSC and NSD (%) when varying the fraction of training data across different segmentation methods. Best results are in **bold**, and second-best are underlined.

Method	10% Data		25% Data		50% Data		100% Data	
	DSC ↑	NSD ↑	DSC ↑	NSD ↑	DSC ↑	NSD ↑	DSC ↑	NSD ↑
Unimodal Approaches								
UNet [63]	60.95	64.43	62.74	66.16	71.61	75.14	78.49	82.07
UNet++ [85]	63.72	67.08	65.86	69.21	73.15	76.31	78.44	81.79
DeepLabv3 [12]	61.32	64.84	65.39	69.10	68.58	72.57	73.28	77.42
AttnUNet [56]	62.78	66.25	64.97	68.53	71.34	74.96	76.30	79.77
nnUNet [34]	73.45	77.37	76.73	80.66	78.86	82.68	81.40	85.08
Swin-UNet [9]	53.04	57.91	54.69	59.24	55.89	61.25	65.03	69.32
TransUNet [11]	52.69	56.38	55.25	58.95	55.22	59.30	67.22	71.15
Generic Text-driven Approaches								
LViT [50]	66.51	68.80	75.66	78.12	78.88	81.34	83.35	85.89
Ariadne's Thread [81]	61.34	62.75	63.09	64.51	65.65	66.92	70.07	71.49
CLIP-Based Approaches								
EoMT-CLIP [41]	74.07	77.41	76.29	79.84	79.19	82.78	82.93	86.35
CLIPSeg [51]	74.66	77.75	78.31	81.34	79.63	82.58	84.87	87.74
DenseCLIP [61]	67.84	70.33	70.23	72.70	72.09	74.45	74.19	76.89
ZegCLIP [86]	61.25	63.72	72.46	75.01	76.21	78.80	78.98	81.69
SAN [72]	74.13	76.97	76.13	78.91	78.80	81.52	81.62	84.35
MaPLe [42]	66.27	68.75	71.53	73.95	74.60	77.12	74.60	77.10
MaPLe [42] + Decoder	74.81	77.90	79.64	82.60	82.81	85.80	84.94	87.91
VLSM-Adapter [19]	74.47	77.50	77.63	80.53	80.83	83.77	83.85	86.72
CausalCLIPSeg [13]	71.19	73.74	75.42	78.00	78.60	81.22	81.34	84.20
CAT-Seg [16]	78.76	81.50	81.12	83.92	83.32	85.61	85.90	88.31
`MedCLIPSeg` (Ours)	**81.10**	**83.94**	**85.08**	**87.85**	**87.18**	**89.95**	**88.66**	**91.35**

Datasets: We experiment on diverse medical imaging datasets spanning six organs and five modalities, covering clinically critical segmentation tasks including *tumor*, *polyp*, and *skin lesion* segmentation, each representing challenging, high-impact applications with ambiguous boundaries and large domain variability. **For supervised settings and data-efficiency tests**, we use BUSI [1], BTMRI [14], ISIC [17, 66], Kvasir-SEG [36], QaTa-COV19 [18], and EUS [35]. **For domain generalization tests**, evaluations are conducted on BUSUC [33], BUSBRA [28], BUID [4], UDIAT [8], CVC-ColonDB [65], CVC-ClinicDB [6], CVC-300 [68], BKAI [55], BRISC [24], and UWaterlooSkinCancer [2, 27], each introducing substantial domain shifts. For datasets with missing text, we generated prompts using GPT-5 [57] and applied image processing techniques to identify different visual attributes for each image. Dataset and prompt details are provided in Appendices A and C.

Table 2. **Domain generalization:** Models are trained on a source dataset and evaluated on OOD target datasets without adaptation. DSC (%) values are reported where the best results are in **bold**, and second-best are underlined.

Method	Breast Ultrasound					Polyp Endoscopy					Brain MRI		Skin Dermatoscopy	
	Source	Target				Source	Target				Source	Target	Source	Target
	BUSI	BUSBRA	BUSUC	BUID	UDIAT	Kvasir-SEG	ColonDB	ClinicDB	CVC300	BKAI	BTMRI	BRISC	ISIC	UWaterloo
LViT [50]	75.32	59.41	67.95	53.51	65.60	85.29	60.01	75.27	70.22	67.17	81.41	71.86	91.21	58.87
CLIPSeg [51]	80.95	63.66	75.03	68.43	56.67	81.98	59.93	71.49	72.74	66.46	86.33	77.61	90.55	80.19
DenseCLIP [61]	71.85	53.34	70.97	63.53	54.93	79.32	56.38	68.08	64.71	61.63	70.30	34.12	89.29	53.39
ZegCLIP [86]	72.08	61.08	73.57	71.75	52.41	78.46	53.46	69.75	60.73	65.60	76.65	66.31	81.45	38.60
SAN [72]	77.99	64.37	74.15	58.13	61.98	83.16	61.82	74.46	80.36	69.31	85.27	71.60	91.39	82.51
MaPLe [42]	66.37	50.08	71.52	70.77	57.81	76.12	48.09	59.64	63.80	56.94	75.40	45.19	88.31	69.12
MaPLe [42] + Decoder	80.49	55.89	64.96	60.66	59.44	83.46	61.53	71.20	74.62	66.93	85.08	71.46	90.10	81.83
VLSM-Adapter [19]	80.90	68.48	82.37	75.26	69.16	85.89	63.51	76.09	75.24	71.59	85.03	68.92	91.30	82.17
CausalCLIPSeg [13]	76.11	55.87	69.12	64.49	48.90	78.77	41.65	57.54	45.77	52.56	81.71	53.96	89.47	48.73
CAT-Seg [16]	81.83	70.94	81.48	73.37	70.30	86.43	68.49	70.35	78.12	74.35	84.86	76.28	91.27	82.02
MedCLIPSeg (Ours)	**85.72**	**75.06**	**84.37**	**78.99**	**74.64**	**90.15**	**71.90**	**80.80**	**80.82**	**79.15**	**88.03**	**80.92**	**92.54**	**83.53**

Implementation Details: We use UniMedCLIP ViT-B/16 [43] with PubMedBERT [76] as the backbone. All models are trained for 100 epochs (10 for EUS) with a learning rate of 3×10^{-4}, batch size 24, and Adam optimizer [44] under cosine annealing. The segmentation loss combines Dice and binary cross-entropy (BCE) losses (equal weights). Experiments are conducted on a single NVIDIA A100 GPU (40GB RAM). All settings were identical across all experiments, and all CLIP-based baselines use the same UniMedCLIP backbone for fair comparison. We use Dice Similarity Coefficient (DSC) and normalized surface distance (NSD) to compare segmentation accuracy.

4.2. Data Efficiency Evaluation

As shown in Table 1, MedCLIPSeg consistently outperforms both unimodal and multimodal baselines, notably the strong CLIP-based CAT-Seg, with **2–3%** gains at 10% data and **3–4%** at 50%. Compared to EoMT-CLIP [41], a variant without the PVL Adapters and $\mathcal{L}_{\text{SoftCon}}$, MedCLIPSeg achieves further **+7.0%** and **+8.8%** DSC improvements at 10% and 25% efficiency, respectively, highlighting the effectiveness of these components for data-efficient learning.

4.3. Domain Generalization

Table 2 reports cross-dataset performance. MedCLIPSeg achieves 85.7% DSC on BUSI, 84.4% on BUSUC, 90.2% on Kvasir-SEG, 88.0% on BTMRI, and 92.5% on ISIC, consistently outperforming SAN [72] and CAT-Seg [16]. Despite substantial distribution shifts, such as lighting/signal gain, zoom, and viewpoint/field-of-view variations in polyp and ultrasound datasets, our model preserves contour fidelity and segmentation quality, demonstrating robust generalization across domains.

4.4. Effectiveness of Key Design Components

Table 3 quantifies the impacts of MedCLIPSeg's components. HM denotes the harmonic mean between the ID and OOD DSC (%) scores. Removing the PVL Adapters causes the largest DSC drop (–7.9% ID, –23.8% OOD), emphasizing their role in robust multi-modal alignment. Replacing probabilistic attention with deterministic attention reduces OOD DSC by 15.9%, confirming the value of uncertainty-aware formulation. Bidirectional interaction further enhances performance, while excluding $\mathcal{L}_{\text{SoftCon}}$ decreases HM DSC by 1.92%, highlighting its positive role in maintaining nuanced cross-modal alignment.

Table 3. **Effect of different key components.**

Method	Domain Generalization		
	ID DSC (%)	OOD DSC (%)	HM DSC (%)
MedCLIPSeg (Ours)	**89.11**	**79.02**	**83.76**
Probabilistic Vision-Language Adapters			
w/o PVL Adapters	$81.23_{(-7.88)}\downarrow$	$55.23_{(-23.79)}\downarrow$	$65.75_{(-18.01)}\downarrow$
w/o Gating	$87.55_{(-1.56)}\downarrow$	$76.79_{(-2.23)}\downarrow$	$81.82_{(-1.94)}\downarrow$
w/o Attn_{PVL}	$86.21_{(-2.90)}\downarrow$	$74.13_{(-4.89)}\downarrow$	$79.71_{(-4.05)}\downarrow$
Deterministic MedCLIPSeg	$87.68_{(-1.43)}\downarrow$	$63.12_{(-15.90)}\downarrow$	$73.40_{(-10.36)}\downarrow$
Bidirectional Multimodal Interaction			
w/o Visual Adaptation	$81.50_{(-7.61)}\downarrow$	$64.40_{(-14.62)}\downarrow$	$71.95_{(-11.81)}\downarrow$
w/o Textual Adaptation	$88.83_{(-0.28)}\downarrow$	$76.40_{(-2.62)}\downarrow$	$82.15_{(-1.61)}\downarrow$
w/o Bidirectional Interaction	$88.71_{(-0.40)}\downarrow$	$77.71_{(-1.31)}\downarrow$	$82.85_{(-0.91)}\downarrow$
Unimodal MedCLIPSeg	$86.53_{(-2.58)}\downarrow$	$73.49_{(-5.53)}\downarrow$	$79.48_{(-4.28)}\downarrow$
Contrastive Loss			
w/o $\mathcal{L}_{\text{SoftCon}}$	$87.24_{(-1.87)}\downarrow$	$77.08_{(-1.94)}\downarrow$	$81.84_{(-1.92)}\downarrow$
Hard Targets	$88.34_{(-0.77)}\downarrow$	$77.64_{(-1.38)}\downarrow$	$82.65_{(-1.11)}\downarrow$
Attention-pooled $\mathcal{L}_{\text{SoftCon}}$	$88.73_{(-0.38)}\downarrow$	$75.60_{(-3.42)}\downarrow$	$81.64_{(-2.12)}\downarrow$

Table 4. **Effect of text prompt design.**

Text Prompt Design	ID DSC (%)	OOD DSC (%)	HM DSC (%)
Contradictory	68.60	63.21	65.79
Missing Location	86.98	77.75	82.11
Overdescriptive	82.93	74.49	78.48
Underdescriptive	66.91	49.38	56.82
Original	**89.11**	**79.02**	**83.76**

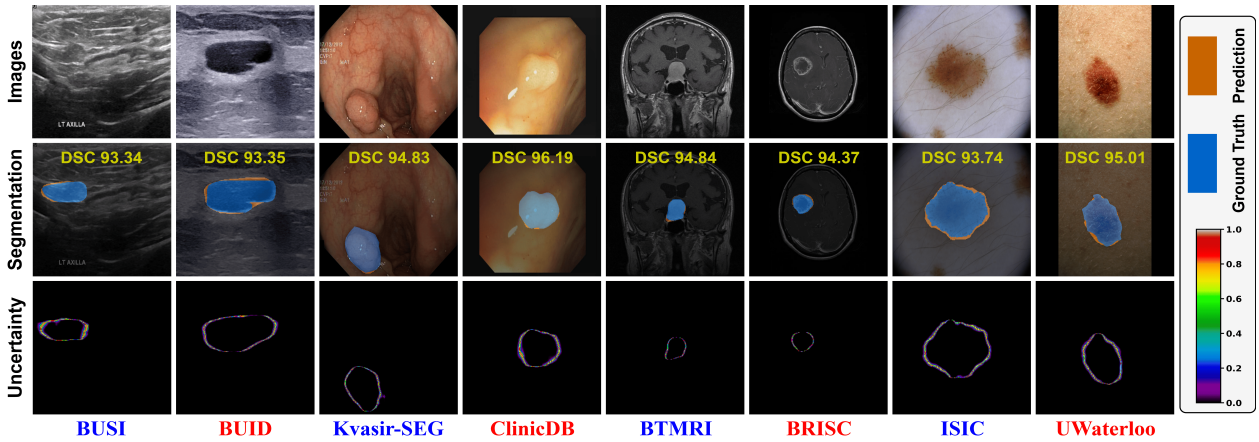


Figure 4. **Segmentation and uncertainty visualizations.** Uncertainty peaks along lesion boundaries and remains consistent across diverse datasets, indicating reliable calibration and generalization. ID data are in **blue** while OOD data are in **red**.

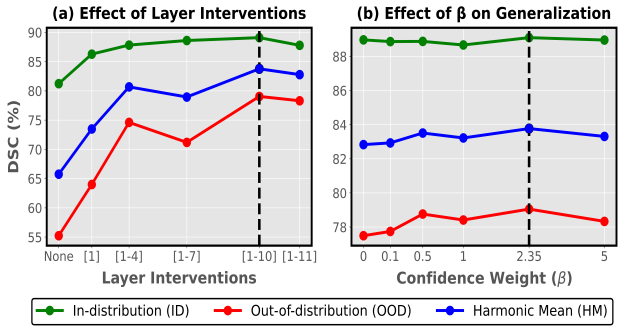


Figure 5. Layer-wise interventions (***left***) and confidence weighting (β) (***right***) ablations averaged on ID and OOD data.

Table 5. **Effect of pre-trained vision–language models.**

Pre-trained Model	ID DSC (%)	OOD DSC (%)	HM DSC (%)
CLIP [60]	88.48	74.81	81.07
PubMedCLIP [22]	86.67	73.05	79.28
BiomedCLIP [76]	88.70	77.08	82.48
UniMedCLIP [43]	**89.11**	**79.02**	**83.76**

4.5. Ablation Studies

Layer Interventions: Figure 5(a) shows that deeper layer interventions with `PVL Adapters` steadily improve both ID and OOD segmentation, peaking at Layer 10 (HM DSC 83.76%), with a slight drop at the final layer.

Confidence Weight (β) Choice: Figure 5(b) shows that β=2.35 yields the best HM DSC from a range of tested values in [0,5], balancing in-domain stability and OOD generalization by effectively calibrating probabilistic attention.

Text Prompt Style: Table 4 shows that concise yet informative prompts (i.e., `Original`) perform best. Removing spatial cues (i.e., location) or adding verbosity lowers performance by 1.65% and 5.28%, respectively. `Contradictory`, which contain internally inconsistent descriptions that can confuse the model, along with `Underspecified` prompts, deteriorate DSCs to 65.79% and 56.82%, respectively (see Appendix E for more details).

CLIP Backbone: Table 5 shows that the backbone choice strongly impacts generalization. Models benefit from stronger and larger-scale CLIP pretraining, with UniMedCLIP [43] providing the most robust transferability.

4.6. Uncertainty Visualization and Reliability

Figure 4 illustrates segmentation and uncertainty maps for both ID and OOD data, where uncertainty consistently concentrates along anatomical boundaries and regions prone to expert disagreement. All subsequent quantitative analyses are computed over **foreground regions** in ID and OOD data. Predicted uncertainty closely follows segmentation errors, achieving strong Spearman correlations of **(87.57%, 80.41%)** for (ID, OOD), respectively. Furthermore, probabilistic modeling improves calibration, reducing the Brier scores from **(23.9%, 25.3%)** in the deterministic baseline to **(11.1%, 11.8%)**, as shown in Fig. 1, alleviating overconfidence and enhancing reliability in clinical decision-making.

5. Conclusion

We presented `MedCLIPSeg`, a probabilistic framework for text-driven medical image segmentation by adopting the novel `PVL Adapter` that enables confidence-weighted attention and explicit uncertainty estimation. Leveraging CLIP's pretrained features through bidirectional fusion and a soft patch-level contrastive loss, `MedCLIPSeg` achieves SOTA segmentation performance with high data efficiency, strong out-of-distribution generalization, and well-calibrated uncertainty across six organs and five modalities, advancing reliable VLM for medical AI.

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